

# Circular Economy Ontology Network (CEON) - Resource ODP

## Metadata

**IRI**

<http://w3id.org/CEON/ontology/resourceODP/>

**Title**

Circular Economy Ontology Network (CEON) - Resource ODP

**Publisher**

Onto-DESIDE

**Creator**

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**Contributor**

Eva Blomqvist

**Date Created**

2025-03-20

**Date Issued**

2025-06-30

**License**

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**Version Iri**

<http://w3id.org/CEON/ontology/resourceODP/0.5/>

**Version Info**

0.5

**Prior Version**

0.4

**Preferred Namespace Prefix**

ceon-resourceODP

**Preferred Namespace Uri**

<http://w3id.org/CEON/ontology/resourceODP/>

**Description**

A core ODP of the CEON ontology network defining aspects of the resource concept.

**Covers Requirements**

Covers the following requirements from Onto-DESIDE D3.1: CVN-Resource-1,3, CVN-Composition-1,2, CVN-ResrouceType-4, C7-3, E2-2, E4-6, E5-1, E6-3, T3-1.

Covers the following requirements from Onto-DESIDE D3.2: CE1-2, CE3-3, CE4-3, CE5-2, CE9-1, CE10-1, CE11-8, CE12-2, CE12-4, C7-3, E2-2, E4-6, E5-1, E6-3, T3-1, T19-1, T20-1, T22-2.

## Classes

|                       |                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------|
| Location <sup>c</sup> |                                                                                                               |
| <hr/>                 |                                                                                                               |
| IRI                   | <a href="http://w3id.org/CEON/ontology/location/Location">http://w3id.org/CEON/ontology/location/Location</a> |
| In Range Of           | <a href="#">hasResourceLocation</a> <sup>op</sup>                                                             |

## Asset<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/Asset>

**Super Class Of**  
[Information<sup>C</sup>](#)  
[Resource<sup>C</sup>](#)

## Batch Of Objects<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/BatchOf0bjects>

**Sub Class Of**  
[Resource<sup>C</sup>](#)  
[hasPhysicalObject<sup>op</sup>](#) only [PhysicalObject<sup>C</sup>](#) *and* [hasPhysicalObject<sup>op</sup>](#) some [PhysicalObject<sup>C</sup>](#)

**In Domain Of**  
[batchID<sup>dp</sup>](#)  
[batchLabel<sup>dp</sup>](#)  
[batchSize<sup>dp</sup>](#)

**In Range Of** [hasBatch<sup>op</sup>](#)

**Restriction** [batchSize<sup>dp</sup>](#) *exactly* 1 [BatchOfObjects<sup>C</sup>](#)

## Constituent<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/Constituent>

**In Range Of** [hasConstituent<sup>op</sup>](#)

## Digital Object<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/Digital0bject>

**Sub Class Of** [Resource<sup>C</sup>](#)

## Information<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/Information>

**Sub Class Of** [Asset<sup>C</sup>](#)

**In Domain Of**  
[containsInformation<sup>op</sup>](#)  
[isAbout<sup>op</sup>](#)

**In Range Of**  
[containsInformation<sup>op</sup>](#)  
[isRealizationOf<sup>op</sup>](#)

## Matter<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/Matter>

**In Range Of** [hasMatter<sup>op</sup>](#)

## Physical Object<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/PhysicalObject>

**Sub Class Of** [Resource<sup>C</sup>](#)  
[hasConstituent<sup>op</sup>](#) only [Constituent<sup>C</sup>](#) *and* [hasConstituent<sup>op</sup>](#) some [Constituent<sup>C</sup>](#)  
[hasMatter<sup>op</sup>](#) some [Matter<sup>C</sup>](#) *and* [hasMatter<sup>op</sup>](#) only [Matter<sup>C</sup>](#)

**In Domain Of** [hasConstituent<sup>op</sup>](#)  
[hasMatter<sup>op</sup>](#)

**In Range Of** [hasPhysicalObject<sup>op</sup>](#)

## Resource<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/Resource>

**Is Defined By** ISO 59004:2024 - 3.1.5 resource

**Sub Class Of** [Asset<sup>C</sup>](#)  
[hasResourceCondition<sup>op</sup>](#) some [ResourceCondition<sup>C</sup>](#) *and* [hasResourceCondition<sup>op</sup>](#) only [ResourceCondition<sup>C</sup>](#)  
[hasResourceProperty<sup>op</sup>](#) some [ResourceProperty<sup>C</sup>](#) *and* [hasResourceProperty<sup>op</sup>](#) only [ResourceProperty<sup>C</sup>](#)  
[hasResourceQuality<sup>op</sup>](#) only [ResourceQuality<sup>C</sup>](#) *and* [hasResourceQuality<sup>op</sup>](#) some [ResourceQuality<sup>C</sup>](#)

**In Domain Of** [hasResourceCondition<sup>op</sup>](#)  
[hasResourceLocation<sup>op</sup>](#)  
[hasResourceProperty<sup>op</sup>](#)  
[hasResourceQuality<sup>op</sup>](#)

**Super Class Of** [BatchOfObjects<sup>C</sup>](#)  
[DigitalObject<sup>C</sup>](#)  
[PhysicalObject<sup>C</sup>](#)  
[SetOfObjects<sup>C</sup>](#)

## Resource Condition<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/ResourceCondition>

**In Range Of** [hasResourceCondition<sup>op</sup>](#)

## Resource Property<sup>C</sup>

|                           |                                                                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                | <a href="http://w3id.org/CE0N/ontology/resourceODP/ResourceProperty">http://w3id.org/CE0N/ontology/resourceODP/ResourceProperty</a> |
| <b><u>In Range Of</u></b> | <a href="#">hasResourceProperty</a> <sup>op</sup>                                                                                   |

## Resource Quality<sup>C</sup>

|                           |                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                | <a href="http://w3id.org/CE0N/ontology/resourceODP/ResourceQuality">http://w3id.org/CE0N/ontology/resourceODP/ResourceQuality</a> |
| <b><u>In Range Of</u></b> | <a href="#">hasResourceQuality</a> <sup>op</sup>                                                                                  |

## Set Of Objects<sup>C</sup>

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/resourceODP/SetOf0bjects">http://w3id.org/CE0N/ontology/resourceODP/SetOf0bjects</a>                                                                                                                                                                                                                                                                                                           |
| <b><u>Sub Class Of</u></b> | <a href="#">Resource</a> <sup>C</sup><br><a href="#">hasBatch</a> <sup>op</sup> some <a href="#">BatchOfObjects</a> <sup>C</sup> and <a href="#">hasBatch</a> <sup>op</sup> only <a href="#">BatchOfObjects</a> <sup>C</sup><br><a href="#">hasPhysicalObject</a> <sup>op</sup> only <a href="#">PhysicalObject</a> <sup>C</sup> and <a href="#">hasPhysicalObject</a> <sup>op</sup> some <a href="#">PhysicalObject</a> <sup>C</sup> |
| <b><u>In Domain Of</u></b> | <a href="#">hasBatch</a> <sup>op</sup>                                                                                                                                                                                                                                                                                                                                                                                                |

## Object Properties

### has location<sup>op</sup>

|                                 |                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                      | <a href="http://w3id.org/CE0N/ontology/location/hasLocation">http://w3id.org/CE0N/ontology/location/hasLocation</a> |
| <b><u>Super Property Of</u></b> | <a href="#">hasResourceLocation</a> <sup>op</sup>                                                                   |

### contains information<sup>op</sup>

|                               |                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                    | <a href="http://w3id.org/CE0N/ontology/resourceODP/containsInformation">http://w3id.org/CE0N/ontology/resourceODP/containsInformation</a> |
| <b><u>Sub Property Of</u></b> | <a href="#">hasPart</a> <sup>op</sup>                                                                                                     |
| <b><u>Domain</u></b>          | <a href="#">Information</a> <sup>C</sup>                                                                                                  |
| <b><u>Range</u></b>           | <a href="#">Information</a> <sup>C</sup>                                                                                                  |

### has batch<sup>op</sup>

|                      |                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>           | <a href="http://w3id.org/CE0N/ontology/resourceODP/hasBatch">http://w3id.org/CE0N/ontology/resourceODP/hasBatch</a> |
| <b><u>Domain</u></b> | <a href="#">SetOfObjects</a> <sup>C</sup>                                                                           |
| <b><u>Range</u></b>  | <a href="#">BatchOfObjects</a> <sup>C</sup>                                                                         |

## has constituent<sup>op</sup>

|                        |                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasConstituent">http://w3id.org/CE0N/ontology/resource0DP/hasConstituent</a> |
| <b>Sub Property Of</b> | <a href="#">hasPart<sup>op</sup></a>                                                                                            |
| <b>Domain</b>          | <a href="#">PhysicalObject<sup>c</sup></a>                                                                                      |
| <b>Range</b>           | <a href="#">Constituent<sup>c</sup></a>                                                                                         |

## has matter<sup>op</sup>

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasMatter">http://w3id.org/CE0N/ontology/resource0DP/hasMatter</a> |
| <b>Domain</b> | <a href="#">PhysicalObject<sup>c</sup></a>                                                                            |
| <b>Range</b>  | <a href="#">Matter<sup>c</sup></a>                                                                                    |

## has part<sup>op</sup>

|                          |                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>               | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasPart">http://w3id.org/CE0N/ontology/resource0DP/hasPart</a>                                        |
| <b>Super Property Of</b> | <ul style="list-style-type: none"><li>• <a href="#">containsInformation<sup>op</sup></a></li><li>• <a href="#">hasConstituent<sup>op</sup></a></li></ul> |

## has physical object<sup>op</sup>

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasPhysicalObject">http://w3id.org/CE0N/ontology/resource0DP/hasPhysicalObject</a> |
| <b>Domain</b> | <a href="#">SetOfObjects<sup>c</sup></a> or <a href="#">BatchOfObjects<sup>c</sup></a>                                                |
| <b>Range</b>  | <a href="#">PhysicalObject<sup>c</sup></a>                                                                                            |

## has resource condition<sup>op</sup>

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasResourceCondition">http://w3id.org/CE0N/ontology/resource0DP/hasResourceCondition</a> |
| <b>Domain</b> | <a href="#">Resource<sup>c</sup></a>                                                                                                        |
| <b>Range</b>  | <a href="#">ResourceCondition<sup>c</sup></a>                                                                                               |

## has resource location<sup>op</sup>

|                        |                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasResourceLocation">http://w3id.org/CE0N/ontology/resource0DP/hasResourceLocation</a> |
| <b>Sub Property Of</b> | <a href="#">location:hasLocation<sup>op</sup></a>                                                                                         |
| <b>Domain</b>          | <a href="#">Resource<sup>c</sup></a>                                                                                                      |
| <b>Range</b>           | <a href="#">location:Location<sup>c</sup></a>                                                                                             |

### has resource property<sup>op</sup>

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasResourceProperty">http://w3id.org/CE0N/ontology/resource0DP/hasResourceProperty</a> |
| <b>Domain</b> | <a href="#">Resource</a> <sup>c</sup>                                                                                                     |
| <b>Range</b>  | <a href="#">ResourceProperty</a> <sup>c</sup>                                                                                             |

### has resource quality<sup>op</sup>

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/hasResourceQuality">http://w3id.org/CE0N/ontology/resource0DP/hasResourceQuality</a> |
| <b>Domain</b> | <a href="#">Resource</a> <sup>c</sup>                                                                                                   |
| <b>Range</b>  | <a href="#">ResourceQuality</a> <sup>c</sup>                                                                                            |

### is about<sup>op</sup>

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/isAbout">http://w3id.org/CE0N/ontology/resource0DP/isAbout</a> |
| <b>Domain</b> | <a href="#">Information</a> <sup>c</sup>                                                                          |

### is realization of<sup>op</sup>

|              |                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>   | <a href="http://w3id.org/CE0N/ontology/resource0DP/isRealizationOf">http://w3id.org/CE0N/ontology/resource0DP/isRealizationOf</a> |
| <b>Range</b> | <a href="#">Information</a> <sup>c</sup>                                                                                          |

## Datatype Properties

### batch id<sup>dp</sup>

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/batchID">http://w3id.org/CE0N/ontology/resource0DP/batchID</a> |
| <b>Domain</b> | <a href="#">BatchOfObjects</a> <sup>c</sup>                                                                       |
| <b>Range</b>  | <a href="#">xsd:string</a>                                                                                        |

### batch label<sup>dp</sup>

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/resource0DP/batchLabel">http://w3id.org/CE0N/ontology/resource0DP/batchLabel</a> |
| <b>Domain</b> | <a href="#">BatchOfObjects</a> <sup>c</sup>                                                                             |
| <b>Range</b>  | <a href="#">xsd:string</a>                                                                                              |

batch size<sup>dp</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/batchSize>

**Domain** [BatchOfObjects<sup>c</sup>](#)

**Range** [xsd:nonNegativeInteger](#)

## Annotation Properties

contributor<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/contributor>

created<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/created>

creator<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/creator>

description<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/description>

issued<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/issued>

license<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/license>

publisher<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/publisher>

title<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/title>

preferred namespace prefix<sup>ap</sup>

**IRI** <http://purl.org/vocab/vann/preferredNamespacePrefix>

|                                       |                                                                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| preferred namespace uri <sup>ap</sup> |                                                                                                                 |
| IRI                                   | <a href="http://purl.org/vocab/vann/preferredNamespaceUri">http://purl.org/vocab/vann/preferredNamespaceUri</a> |

  

|                                   |                                                                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| covers requirements <sup>ap</sup> |                                                                                                                                                                                               |
| IRI                               | <a href="http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements">http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements</a> |

## Namespaces

|          |                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| :        | <a href="http://w3id.org/CEON/ontology/resourceODP/">http://w3id.org/CEON/ontology/resourceODP/</a>                                                     |
| dcterms  | <a href="http://purl.org/dc/terms/">http://purl.org/dc/terms/</a>                                                                                       |
| geo      | <a href="http://www.opengis.net/ont/geosparql#">http://www.opengis.net/ont/geosparql#</a>                                                               |
| location | <a href="http://w3id.org/CEON/ontology/location/">http://w3id.org/CEON/ontology/location/</a>                                                           |
| odp      | <a href="http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#">http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#</a> |
| owl      | <a href="http://www.w3.org/2002/07/owl#">http://www.w3.org/2002/07/owl#</a>                                                                             |
| prov     | <a href="http://www.w3.org/ns/prov#">http://www.w3.org/ns/prov#</a>                                                                                     |
| rdf      | <a href="http://www.w3.org/1999/02/22-rdf-syntax-ns#">http://www.w3.org/1999/02/22-rdf-syntax-ns#</a>                                                   |
| rdfs     | <a href="http://www.w3.org/2000/01/rdf-schema#">http://www.w3.org/2000/01/rdf-schema#</a>                                                               |
| vann     | <a href="http://purl.org/vocab/vann/">http://purl.org/vocab/vann/</a>                                                                                   |
| xsd      | <a href="http://www.w3.org/2001/XMLSchema#">http://www.w3.org/2001/XMLSchema#</a>                                                                       |

## Legend

|    |                       |
|----|-----------------------|
| c  | Classes               |
| op | Object Properties     |
| dp | Datatype Properties   |
| ap | Annotation Properties |